

Automated Data Annotation Strategy: Foundation Model Distillation for Arthropod Detection

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February 15, 2026

1 Executive Summary

To train a robust YOLOv8 object detector on our dataset of approximately 50,000 high-resolution arthropod images, we require bounding box annotations. Manual annotation of a dataset of this magnitude is estimated to require over 500 human-hours.

We propose an **Automated Labeling Pipeline** utilizing *Zero-Shot Object Detection*. We will leverage a heavy-weight Foundation Model (Grounding DINO) to act as a “Teacher,” generating labels which will then be used to train a lightweight “Student” model (YOLOv8) for mobile deployment.

2 Theoretical Framework: Knowledge Distillation

The core theoretical concept driving this strategy is **Knowledge Distillation** (specifically, Label Distillation). We perform a transfer of inductive biases from a large, open-vocabulary model to a closed-vocabulary, real-time model.

2.1 The Teacher: Grounding DINO

Grounding DINO is an open-set object detector that fuses text and image modalities. Unlike traditional detectors trained on a fixed set of classes (e.g., COCO’s 80 classes), Grounding DINO learns a joint embedding space for language and vision.

Given an image I and a text prompt T (e.g., “insect”), the model predicts bounding boxes B by maximizing the similarity between the image region features and the text embedding:

$$B_{pred} = \underset{B}{\operatorname{argmax}} P(B|I, T) \quad (1)$$

This allows us to label classes the model has never explicitly been trained on, provided it understands the semantic concept of the class.

Note on Label Noise: We acknowledge that teacher-generated labels are inherently noisy compared to human verification. However, empirical evidence in semi-supervised learning suggests that YOLO-style detectors are robust to moderate label noise when trained at scale, as the network tends to learn the dominant signal (the insect) over idiosyncratic errors (random noise).

2.2 The Student: YOLOv8

YOLO (You Only Look Once) is optimized for inference speed. It requires fixed class definitions and cannot process text prompts. By training YOLO on the output of Grounding DINO, we effectively “compile” the massive knowledge of the foundation model into a compact numerical representation suitable for mobile deployment.

2.3 Handling Class Imbalance

Given the disparate species counts in our dataset (e.g., 50k Beetles vs. 600 Ticks), simple random sampling during training will lead to bias. During the YOLO training phase, class imbalance will be

mitigated via oversampling of minority classes, focal loss to down-weight easy examples, or class-weighted loss functions as needed.

3 Implementation Architecture on Azure

To optimize for cost and throughput, we utilize the “Unzip-on-Compute” pattern rather than streaming small files from cloud storage.

3.1 Infrastructure Components

- **Storage Layer:** Azure Blob Storage (`raw-data-zips`). Contains the 12 archival ZIP files (approx. 60GB).
- **Compute Layer:** Azure Machine Learning Compute Instance (`Standard_NC4as_T4_v3`). Equipped with 4 vCPUs, 28GB RAM, and 1x NVIDIA Tesla T4 GPU.
- **Orchestration:** Python script utilizing the `autodistill` library.

3.2 Data Flow Pipeline

1. **Ingestion:** The Compute Instance downloads the target ZIP file (e.g., `BugLiteAI_spiders.zip`) from the Blob Container to its high-speed local NVMe SSD.
2. **Extraction:** The archive is unzipped locally. This avoids the high latency of listing thousands of blobs over the network.
3. **Inference (The Labeling Loop):**
 - The Grounding DINO model is loaded onto the T4 GPU.
 - **Ontology Definition:** We map visual concepts to a single detection class. For detection purposes, visually similar arthropods (e.g., spiders, scorpions) are mapped to a single coarse ‘insect’ class to improve recall.
 - **Prompt Strategy:** We utilize a broad prompt ontology: `{"insect", "spider", "bug", "beetle", "caterpillar"}`. This ensures maximum recall across diverse morphologies.
 - **Inference:** The model generates bounding box proposals based on the text prompts.
4. **Post-Processing (The “Top-1” Logic):** Crucially, we do not accept all predictions. We apply a strict heuristic derived from our Pilot findings (see Section 5):
 - **Confidence Ranking:** Grounding DINO assigns a logit score (confidence) to each box based on the dot-product similarity between the visual feature map and the text embedding.
 - **Selection Rule:** We select **only the single highest-confidence box** per image.

$$B_{final} = \operatorname{argmax}_{b \in B_{pred}} \operatorname{Score}(b) \quad (2)$$

- This implicitly filters out background clutter (flowers, shadows) which typically have lower confidence scores than the primary subject.
5. **Serialization:** The single accepted bounding box is written to a text file in the standard YOLO format.
 6. **Egress:** The resulting folder of text files is zipped and uploaded to a new Azure Blob Container: `labeled-data-yolo`.

4 Pilot Phase Strategy

Before executing the full pipeline on 50,000 images, we executed a **Pilot Test** to validate the infrastructure and estimate costs.

- **Sampling Rate (α):** We utilized a sampling parameter $\alpha = 0.05$ (5%), processing approximately 2,500 images.
- **Cost Projection:** Based on pilot logs, the model processes images at ≈ 2.7 seconds/image. The estimated compute cost for the full run is $< \$25$.

5 Pilot Findings and Refinement

During the visual inspection of the pilot data, we identified two specific artifacts that necessitated the “Top-1” logic described in Section 3.2.

5.1 The Multi-Box Synonyms Issue

Because our ontology includes synonyms (e.g., “insect” and “bug”), Grounding DINO often detects the same object multiple times, stacking identical boxes on top of each other (one box for “insect”, one for “bug”).

- **NMS Limitation:** While Non-Maximum Suppression (NMS) can merge these based on Intersection over Union (IoU), it creates processing overhead and does not solve the issue of distinct false positives.

5.2 The “Flower vs. Insect” Distractor Issue

In approximately 15% of pilot images, the model detected the insect correctly but *also* placed bounding boxes around high-contrast background elements, such as flowers or dark spots on leaves.

- **Finding:** NMS cannot remove these boxes because they do not overlap geometrically with the insect box.
- **Resolution via Confidence:** We observed that the actual insect consistently yielded a higher confidence score (> 0.60) compared to the distractors ($0.35 - 0.45$).
- **Justification for Top-1:** Since iNaturalist “Research Grade” images are curated to contain a single subject of interest, we can safely assume that the highest confidence detection corresponds to the target arthropod. Enforcing a Top-1 rule eliminates both the synonym stacking and the background distractors in a single operation.

6 Human-in-the-Loop Verification

To ensure quality control, we apply **Stratified Sampling** on the final dataset.

1. Randomly sample $N = 100$ images from each class after auto-labeling.
2. Load these samples into a verification tool (LabelImg/Roboflow).
3. **Metric:** Calculate the IoU of the Top-1 auto-label against a human ground truth.
4. If the IoU < 0.5 for any class, we will refine the prompts for that specific morphology.

7 Conclusion

This architecture minimizes cloud costs by utilizing transient compute and leverages the semantic understanding of Grounding DINO to clean noisy data. By implementing the Top-1 selection heuristic, we effectively convert a generic zero-shot detector into a high-precision single-object cropper suitable for our pipeline.